

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour
Total Marks:30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

192511347

I Shanmuganathan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Sc
Signature of the student:

18 + 18 + 22 + 17 + 12
= 87
12

2) Propositional Logic Representation

- * P : Meena attended all classes
- * Q : Meena gets attendance Credit
- * Statement 1: $P \rightarrow Q$
- * Statement 2: P
- * Goal: Q

3) CNF Conversion & Negated Goal

- * Clause 1 (from Statement): $\neg P \vee Q$
- * Clause 2 (from Statement): P
- * Clause 3 (Negated Goal): $\neg Q$

4) Resolution Steps

- * Step 1: Resolve Clause 1 ($\neg P \vee Q$) and Clause 2 (P) by Canceling Out P and $\neg P$
 - * yields: Clause 4: Q
- * Step 2: Resolve Clause 4 (Q) and Clause 3 ($\neg Q$)

1) Propositional Logic Representation

- * P: Surash has a valid license
- * Q: Surash can rent a car
- * Statement 1: $P \rightarrow Q$
- * Statement 2: P
- * Goal: Q

2) CNF Conversion & Negated Goal:

- * Clause 1: $\neg P \vee Q$
- * Clause 2: P
- * Clause 3: (Negated Goal): $\neg Q$

Resolution Steps:

- * Step 1: Resolve Clause 1 ($\neg P \vee Q$) and Clause 2 (P)
- * Yield Clause 4: Q

- * Step 2: Resolve Clause 4 (Q) and Clause 3 ($\neg Q$)
- * Yield $\square$ (Empty Clause)

Empty Clause:

($\square$)

The goal is Proved

1) Propositional Logic representation

- * P: Pivya scored above 90 percent
- * Q: Pivya wins a gold medal
- * Statement 1: $P \rightarrow Q$
- * Statement 2: P
- * Goal: Q

2) CNF Conversion & Negated Goal

- * Clause 1: $\neg P \vee Q$
- * Clause 2: P
- * Clause 3: $\neg Q$

3) Resolution Steps:

- * Step 1: Resolve clause 1 ($\neg P \vee Q$) and clause 2 (P)
 - * yield clause 4: Q
- * Step 2: Resolve clause 4 (Q) and clause 3 ($\neg Q$)
 - * yield ($\square$) (Empty clause)

4) Empty clause

($\square$)

1) Propositional Logic representation:

- * P : Farhan posed for interview
- * Q : Farhan is hired
- * Statement 1 : $P \rightarrow Q$
- * Statement 2 : P
- * Goal : Q

2) CNF Conversion & Negated Goal

- * Clause 1 : $\neg P \vee Q$
- * Clause 2 : P
- * Clause 3 : $\neg Q$

3) Resolving Steps

- * Step 1 : Resolve Clause ($\neg P \vee Q$) and Clause 2 (P)
 - * yield Clause 4 : Q
- * Step 2 : Resolve clause 4 (Q) and clause ($\neg Q$)
 - * yield empty clause

4) Empty Clause

($\square$)

The goal is Proved.

- * P : The employee finishes problem
- * Q : The employee becomes permanent staff
- * R : The employee gets full benefit
- * Statement 1: $P \rightarrow Q$
- * Statement 2: $Q \rightarrow R$
- * Statement 3: P
- * Goal R

2) CNF Conversion $\Rightarrow$ Negated Goal

- * Clause 1: $\neg P \vee Q$
- * Clause 2: $\neg Q \vee R$
- * Clause 3: P
- * Clause 4: $\neg R$

3) Resolution Steps $\Rightarrow$ Final Conclusion

- * Step 1: Resolve Clause 1 ($\neg P \vee Q$) and clause 3 (P) by eliminating the complementary literals P and $\neg P$

* yield Clause 5: Q

- * Resolve clause 2 ($\neg Q \vee R$) and Clause 5 (Q)

Eliminating by complementary literals (Q) and $\neg Q$

* yield

* Step 3 Resolve Claus 6 (R) and Claus 4 (7R) by eliminating the complementary literals R and 7R

* yields $\square$ (Empty Claus)

4) Final Conclusion:

Since the Contradiction leads to the derivation of the Empty Claus ($\square$). The negated goal is false

$\therefore$ The original hypothesis holds true. The employee gets full benefits.